Lecture 7

Continuous functions and the derivative

Uses.

Summary 2: A nonempty bounded-above real set has a supremum. An injective function has an inverse on its image.
Summary 3: Convergent real sequences are bounded and have unique limits.
Summary 4: Every bounded real sequence has a convergent subsequence. Closed-interval bounds survive a limit.
Summary 5: Continuity is relative to a domain and preserves sequence limits. Function limits obey the arithmetic laws and are locally bounded.
Summary 6: A continuous image of a real interval is an interval. The standard elementary limits are available without differentiation.

Objectives. *Continuity on a closed interval controls the whole set of function values. We prove that the largest and smallest values are attained and that a continuous strictly monotone function has a continuous inverse. A derivative then describes the local change of a function by one linear term. We distinguish this stronger property from continuity.*

By the end of this lecture you will be able to:

- Prove attainment of extreme values for continuous functions on closed bounded intervals.
- Justify continuity of the inverse of a continuous strictly monotone interval function.
- Calculate derivatives from difference quotients and distinguish one-sided derivatives.
- Express differentiability as a linear approximation with a controlled remainder.

7.1 Extreme values on a closed interval

For a real function $f : A \to \mathbb{R}$, a *global maximum* is a value $f(c)$ with $c \in A$ such that $f(x) \le f(c)$ for every $x \in A$. A *global minimum* reverses this inequality. These are the *extreme values* of the function. Boundedness alone gives upper and lower bounds. Attainment requires an actual input producing the bound.

Theorem 7.1 (Extreme value theorem). *Let $a < b$. A continuous function $f : [a, b] \to \mathbb{R}$ is bounded and attains both a global minimum and a global maximum.*

Proof. First we prove boundedness by contradiction. If f were unbounded, then for each $n \in \mathbb{N}$ we could choose $x_n \in [a, b]$ with $|f(x_n)| \ge n$. Bolzano–Weierstrass gives a subsequence

$x_{n_k} \to c \in [a, b]$. Continuity implies $f(x_{n_k}) \to f(c)$. This convergent sequence is bounded, but $|f(x_{n_k})| \geq n_k \geq k$, a contradiction.

Next let $M = \sup f([a, b])$. This exists because the set of values is nonempty and bounded above. The least-upper-bound property supplies, for each $n \in \mathbb{N}$, an input $y_n \in [a, b]$ satisfying

$$M - 1/n < f(y_n) \leq M.$$

Squeeze gives $f(y_n) \to M$. Another use of Bolzano–Weierstrass supplies $y_{n_k} \to d \in [a, b]$. The subsequence of values still tends to M, while continuity makes it tend to $f(d)$. Uniqueness gives $f(d) = M$. Thus the supremum is attained as a maximum.

Finally, the continuous function $-f$ has a maximum by the proved case. If $-f$ is largest at $p \in [a, b]$, then $f(x) \geq f(p)$ for all $x \in [a, b]$. This gives a minimum of f. ∎

For example, on $[-2, 1]$ the function $f(x) = (x + 1)^2$ has minimum zero at $x = -1$. Since $-1 \leq x + 1 \leq 2$, its values are at most 4, and $f(1) = 4$ attains this maximum. The theorem guarantees attainment before a calculation locates the inputs. It does not assert uniqueness. A constant function attains both extreme values at every input.

The domain and continuity hypotheses cannot be removed. The continuous function $1/x$ on $(0, 1]$ is unbounded. The identity on $(0, 1)$ is bounded but attains neither its infimum 0 nor its supremum 1. On the closed interval $[0, 1]$, the function $f(0) = 0$, $f(x) = 1/x$ for $x > 0$ is unbounded. It fails continuity at zero. A closed bounded interval and continuity are separate requirements.

7.2 Continuity of an inverse

Recall from Lecture 6 that a real function on an interval is *strictly increasing* if $x < y$ implies $f(x) < f(y)$. It is *strictly decreasing* if these output inequalities are reversed. Either property is called *strict monotonicity*. It implies injectivity, so the inverse on the image exists. Non-strict monotonicity would permit constant pieces and would not supply an inverse.

The increasing case can be pictured by bracketing c with inputs x_- and x_+ inside the requested input tolerance. Their images lie on opposite sides of $f(c)$. A sufficiently small output tolerance stays between those two images, so the inverse output lies between x_- and x_+. Figure 7.1 illustrates the case where both sides exist.

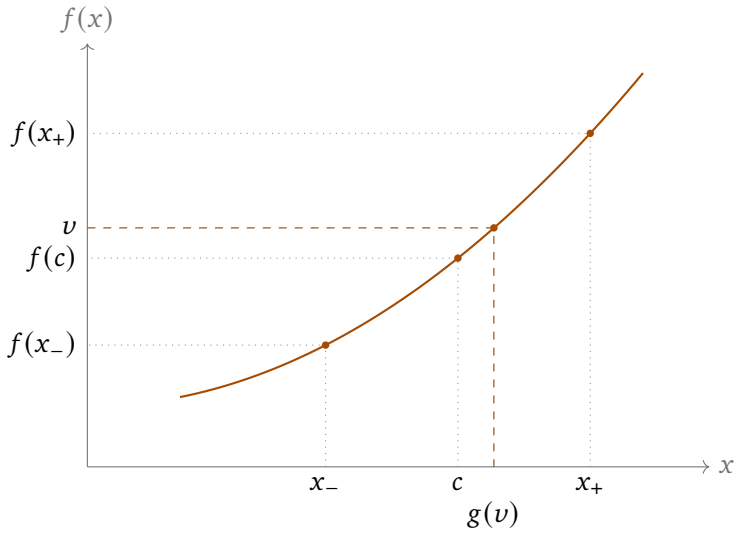


Figure 7.1: For an increasing function at an interior point, an output v between the two bracketing images has inverse $g(v)$ between x_- and x_+. The proof also treats an endpoint, where one side may be absent.

Theorem 7.2 (Continuous inverse on an interval). *Let $I \subseteq \mathbb{R}$ be a nonempty interval, and let $f : I \to \mathbb{R}$ be continuous and strictly monotone. Its image $J = f(I)$ is an interval. The inverse $g : J \to I$ is continuous relative to J.*

Proof. The image is an interval by Lecture 6's intermediate value theorem. Injectivity gives a well-defined inverse on that image. If I has one point, so does J, and relative continuity is automatic. We prove the remaining continuity assertion first for increasing f.

Fix $c \in I$ and a requested input error $\varepsilon > 0$ for the inverse. If I contains points below c, choose $x_- \in I$ with $c - \varepsilon < x_- < c$. Such a point exists by the interval property: between any earlier point and c we may choose a point as close to c as needed. Then $f(c) - f(x_-) > 0$. If I contains points above c, choose $x_+ \in I$ with $c < x_+ < c + \varepsilon$. Then $f(x_+) - f(c) > 0$. Choose $\delta > 0$ no larger than either positive output gap that is present. At least one side is present when I has more than one point.

Take $v \in J$ with $|v - f(c)| < \delta$ and write $x = g(v)$. On a side with x_-, the inequality $v > f(x_-)$ and strict increase imply $x > x_- > c - \varepsilon$. If no point of I lies below c, then $x \geq c$ already. The upper side gives $x < x_+ < c + \varepsilon$ when x_+ was chosen. If that side is absent, $x \leq c$ already. Consequently $|g(v) - g(f(c))| = |x - c| < \varepsilon$. This is continuity of g at the arbitrary point $f(c) \in J$.

For decreasing f, put $F(x) = -f(x)$, which is continuous and strictly increasing. The proved case gives a continuous inverse G on $-J$. The original inverse is $g(v) = G(-v)$. The map $v \mapsto -v$ is continuous, so continuity of composition proves continuity of g on J. ∎

For example, the square map on $[0, \infty)$ is continuous and strictly increasing, and has image $[0, \infty)$ by the school positive-root fact. The theorem gives relative continuity of its inverse $\sqrt{x}$, including at zero. It does not require a two-sided domain at that endpoint.

The same theorem applies to the school restricted sine and tangent maps, and hence to their inverses on their actual images. Derivatives of those inverses will require additional hypotheses in Lecture 8.

7.3 Difference quotients and tangent lines

A point $c \in A \subseteq \mathbb{R}$ is an *interior point* if $(c-\delta, c+\delta) \subseteq A$ for some $\delta > 0$. The derivative at an interior point of any interval can be defined on such an open subinterval. This includes interior points of closed or half-open interval domains.

Let I be an open real interval and let $f : I \to \mathbb{R}$. At $c \in I$, the change in output caused by a nonzero increment h is $f(c+h) - f(c)$. Dividing by the input increment gives the slope of the secant line through $(c, f(c))$ and $(c+h, f(c+h))$. Only increments with $c+h \in I$ are allowed.

Definition 7.3 (Derivative). The function f is *differentiable at* c if the following finite real limit exists. Its value is the *derivative* of f at c:

$$f'(c) = \lim_{h \to 0} \frac{f(c+h) - f(c)}{h}.$$

The function is differentiable on I if it is differentiable at every $c \in I$. The map $c \mapsto f'(c)$ is then its derivative function.

If the derivative exists, the *tangent line* at $(c, f(c))$ is the line of slope $f'(c)$ through that point, given by

$$y = f(c) + f'(c)(x - c).$$

This definition uses a finite limiting slope. An infinite quotient limit is not a derivative in this course's real-valued definition. For position $s(t)$ along a line, the quotient is average velocity over a time increment, and $s'(t)$ is instantaneous velocity when it exists. Its units are the position units divided by the time units.

Figure 7.2 compares the secant and tangent slopes for a square function.

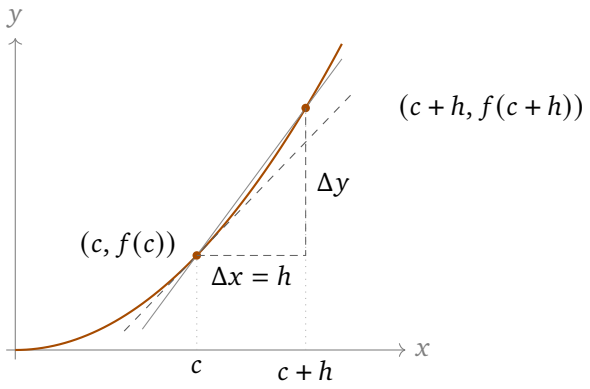


Figure 7.2: The solid grey secant passes through both curve points. The dashed tangent has the limiting secant slope at the fixed base point. The secant slope is $\Delta y / \Delta x$, with $\Delta y = f(c+h) - f(c)$.

Example 7.4 (A derivative from its definition). Let $f(x) = x^2$ on $\mathbb{R}$. At any $c \in \mathbb{R}$, the difference

quotient is

$$\frac{(c+h)^2 - c^2}{h} = 2c + h \longrightarrow 2c.$$

Thus $f'(c) = 2c$. At $c = 1$, the tangent line is $y = 1 + 2(x - 1)$. The identity

$$f(1+h) = 1 + 2h + h^2$$

shows how the tangent approximation differs from the curve at a nearby input. The remainder divided by h is h, which tends to zero.

A constant function has difference quotient zero at every input. The identity function has difference quotient 1. These observations give constant derivative zero and identity derivative 1 without any rule table. For a second direct calculation, take $f(x) = 1/x$ on $(0, \infty)$ at 2. For nonzero h close enough to zero,

$$\frac{f(2+h) - f(2)}{h} = -\frac{1}{2(2+h)} \longrightarrow -\frac{1}{4}.$$

Thus its tangent line there is $y = 1/2 - (x - 2)/4$. The increment must keep $2 + h > 0$, which holds for $|h| < 1$.

7.4 One-sided derivatives and continuity

The *right derivative* uses the same quotient with $h \to 0+$. The *left derivative* uses $h \to 0-$. They are denoted $f'_+(c)$ and $f'_-(c)$ when the corresponding finite limits exist. At an interior point, the derivative exists exactly when both one-sided derivatives exist and agree. Indeed, a two-sided limit has those restrictions. Conversely, the smaller of the two radii for a common requested error controls the full deleted neighbourhood. At an included endpoint of an interval domain, the same one-sided definition uses only the available side.

For example, the function $f(x) = |x|$ has quotient $|h|/h$ at zero. It equals 1 for $h > 0$ and -1 for $h < 0$. The right derivative is 1 and the left derivative is -1, so there is no derivative at zero. Yet $|x|$ is continuous there, because its error from $f(0) = 0$ is exactly $|x|$.

Proposition 7.5 (Differentiability implies continuity). *If f is differentiable at an interior point c of its interval domain, then f is continuous at c.*

Proof. For nonzero h sufficiently close to zero, write

$$f(c+h) - f(c) = h\frac{f(c+h) - f(c)}{h}.$$

The quotient tends to the real number $f'(c)$ and $h \to 0$. The product law gives $f(c+h) - f(c) \to 0$. At $h = 0$ the difference is zero as well. Thus the nearby values tend to the actual value $f(c)$, which is continuity. ∎

The absolute-value example shows that the converse is false. A finite one-sided derivative similarly gives continuity from that side, by the same product identity with the restricted increments. The square root at its domain endpoint zero is continuous, but its right quotient is $\sqrt{h}/h = 1/\sqrt{h}$, which tends to $+\infty$. For every $M > 0$, the choice $0 < h < 1/M^2$ makes this quotient exceed M. There is no finite right derivative there.

7.5 Linear approximation and the differential

For a real function r defined near zero, the notation $r(h) = o(h)$ as $h \to 0$ means $r(h)/h \to 0$ for nonzero h. The notation is read as *little-oh of* h. It specifies the error relative to the size of the increment, not merely that the error tends to zero. For example, $h^2 = o(h)$, whereas $|h|$ is not $o(h)$.

Proposition 7.6 (Linear approximation criterion)**.** *A function f on an open interval containing c is differentiable at c with derivative L exactly when*

$$f(c+h) = f(c) + Lh + r(h), \qquad r(h) = o(h).$$

The remainder is defined at zero by $r(0) = 0$.

Proof. If $f'(c) = L$, define $r(h) = f(c+h) - f(c) - Lh$. For nonzero h, its ratio by h is the difference quotient minus L, which tends to zero. Its value at zero is zero by the defining identity. Conversely, division of the stated representation by a nonzero h gives

$$\frac{f(c+h) - f(c)}{h} = L + \frac{r(h)}{h} \longrightarrow L.$$

This is differentiability with derivative L. ∎

At a fixed base point c, the *differential* is the linear change $dy = f'(c)\,dx$ assigned to an input increment dx. Here dx is a real increment, not a new kind of number. The actual change $\Delta y = f(c+dx) - f(c)$ satisfies $\Delta y = dy + o(dx)$. For each $\varepsilon > 0$, this means that some radius gives $|\Delta y - dy| < \varepsilon |dx|$ for all sufficiently small nonzero dx. It does not assert a universal numerical error bound for a chosen increment before that radius has been established.

For example, approximate $1/2.1$ from the base point 2. The derivative calculated above gives

$$dx = 1/10, \qquad dy = -1/40, \qquad f(2) + dy = \frac{1}{2} - \frac{1}{40} = \frac{19}{40}.$$

For this function the exact remainder can also be computed:

$$\frac{1}{2+h} - \frac{1}{2} + \frac{h}{4} = \frac{h^2}{4(2+h)}.$$

At $h = 1/10$, the remainder is $1/840$. Thus the linear approximation is below the true value $10/21$ by exactly $1/840$. This error statement comes from the exact remainder identity. The derivative alone would give only its limiting relative behaviour.

Summary 7.

- Theorem 7.1: a continuous real function on a closed bounded interval is bounded and attains both extreme values.
- Theorem 7.2: a continuous strictly monotone function on a real interval has an interval image and a continuous inverse on that image.
- Definition 7.3: a derivative is a finite limit of difference quotients. At an interior point, the two one-sided derivatives must agree.
- Proposition 7.5: differentiability implies continuity. The converse fails for $|x|$ at zero.
- Proposition 7.6: differentiability is equivalent to a linear approximation with remainder $o(h)$. The differential is the linear part.

Lecture 8 proves the differentiation rules and derives the elementary derivative table from the standard limits. Sheet 7 practises inverse continuity, derivatives and linear approximation.

References

R. G. Bartle and D. R. Sherbert, *Introduction to Real Analysis*, 4th ed., Wiley, 2011, Chapters 5–6.